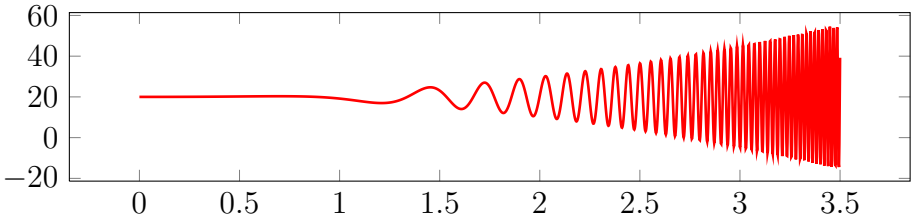
CISS240: Introduction to Programming
Quiz q1007

Name: eagraham1@cougars.ccis.edu Score:

This is a closed-book, no compiler, 5 minute quiz.

Q1. The following is a plot of

$$f(x) = 2x^{2.3}\cos\left(x^{4.5} + 1\right) + 20$$



Write a program that prints the value of x from $a = 1$ to $b = 1.5$ where $f(x)$ reaches a maximum value using a (brute force) search with 1000 equally spaced points on the interval $[1, 1.5]$. The printout must be in fixed point format with 5 decimal places. Skeleton code is provided.

Answer:

```
#include <iostream>
#include <iomanip>
#include <cmath>

int main()
{
    double max_x = 0.0;                 // x where the max of the function occurs
    double max_y = -1000000000.0;       // max value of functions
    double a = 1.0, b = 1.5;            // endpoints
    int n = 1000;                       // number of points
    double dx = (b - a) / n;            // distance between consecutive points

    for (int i = 0; i <= n; ++i)
    {
        double x = a + i * dx;
        double y = 2 * pow(x, 2.3) * cos(pow(x, 4.5) + 1) + 20;)

        if (y > max_y)
```

```
        {
            max_y = y;                // Update max_y
            max_x = x;                // Update max_x
        }
    }
    std::cout << std::fixed << std::setprecision(5)
              << "Max x: " << max_x << std::endl
              << "Max y: " << max_y << std::endl;

    return 0;
}
```

Q2. Write a program that gets a positive integer n from the user and prints the sum of the digits of n. For instance if the user enters 2407, the program prints 13:

```
2407
13
```

Skeleton code is provided.
Answer:

```
#include <iostream>
#include <cmath>

int main()
{
    int n;
    std::cin >> n;

    // compute the number of digits in n and store in num_digits
    int num_digits;

    // compute the sum of digits in n and store in sum_digits
    int sum_digits = 0;

    while(n > 0)
    {
         sum_digits += n % 10;
         n /= 10;
         num_digits++;
    }

    std::cout << sum_digits << '\n';
    return 0;
}
```

Instructions

In the file `thispreamble.tex` look for

```
\renewcommand\AUTHOR{}
```

and enter your email address:

```
\renewcommand\AUTHOR{jdoe5@cougars.ccis.edu}
```

(This is not really necessary since alex will change that for you when you execute `make`.) In your bash shell, execute "`make`" to recompile `main.pdf`. Execute "`make v`" to view `main.pdf`.

Enter your answers in `main.tex`. In the bash shell, execute "`make`" to recompile `main.pdf`. Execute "`make v`" to view `main.pdf`.

For each question, you'll see boxes for you to fill. For small boxes, if you see

```
1 + 1 = \answerbox{}.
```

you do this:

```
1 + 1 = \answerbox{2}.
```

`answerbox` will also appear in "true/false" and "multiple-choice" questions.

For longer answers that need typewriter font, if you see

```
Write a C++ statement that declares an integer variable name x.
\begin{answercode}
\end{answercode}
```

you do this:

```
Write a C++ statement that declares an integer variable name x.
\begin{answercode}
int x;
\end{answercode}
```

`answercode` will appear in questions asking for code, algorithm, and program output. In this case, indentation and spacing is significant. For program output, I do look at spaces and newlines.

For long answers (not in typewriter font) if you see

```
What is the color of the sky?
\begin{answerlong}
\end{answerlong}
```

you can write

```
What is the color of the sky?
\begin{answerlong}
The color of the sky is blue.
\end{answerlong}
```

A question that begins with "T or F or M" requires you to identify whether it is true or false, or meaningless. "Meaningless" means something's wrong with the question and it is not well-defined. Something like "$1 + 2 = 4$" is either true or false (of course it's false). Something like "$1+_2 = 4$?" does not make sense.

When writing results of computations, make sure it's simplified. For instance write 2 instead of $1 + 1$.

Higher level classes.

For students beyond 245: You can put LaTeX commands in `answerlong`.

More examples of meaningless statements: Questions such as "Is $42 = 1+_2$ true or false?" or "Is $42 = \{2\}^{\{3\}}$ true or false?" does not make sense. "Is $P(42) = \{42\}$ true or false?" is meaningless because $P(X)$ is only defined if X is a set. For "Is $1 + 2 + 3$ true or false?", "$1 + 2 + 3$" is well–defined but as a "numerical expression", not as a "proposition", i.e., it cannot be true or false. Therefore "Is $1 + 2 + 3$ true or false?" is also not a well-defined question.

More examples of simplification: When you write down sets, if the answer is $\{1\}$, do not write $\{1, 1\}$. And when the values can be ordered, write the elements of the set in ascending order. When writing polynomials, begin with the highest degree term.

When writing a counterexample, always write the simplest.